

GALATI, Romania

WMO: 153100

Lat: 45.4731N

Lon: 28.0322E

Elev: 70

StdP: 100.48

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.6	-10.1	-15.8	1.0	-11.6	-13.2	1.2	-8.5	10.2	-3.6	9.1	-2.3	3.2	20	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	33.9	21.5	32.3	20.8	30.7	20.3	23.4	29.8	22.4	28.9	21.6	28.0	2.3	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.4	16.2	26.3	20.4	15.2	25.2	19.5	14.3	24.4	69.9	29.8	66.1	28.8	63.0	27.9	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.3	6.4	-16.2	36.5	3.0	1.7	-18.4	37.7	-20.1	38.8	-21.8	39.7	-24.0	41.0		
(5)			-16.5	25.1	3.0	2.0	-18.7	26.5	-20.4	27.7	-22.1	28.8	-24.2	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-1.2	1.3	6.2	12.2	18.0	21.9	24.0	23.7	18.4	12.2	6.6	0.7	(6)
(7)		DBStd	9.78	4.95	5.40	4.75	3.89	3.52	3.17	2.84	3.23	3.69	4.35	4.72	4.89	(7)
(8)		HDD10.0	1191	347	247	136	22	1	0	0	0	0	28	122	289	(8)
(9)		HDD18.3	2844	605	478	378	186	49	7	1	2	44	196	352	547	(9)
(10)		CDD10.0	1941	0	2	17	89	249	356	435	425	252	95	21	1	(10)
(11)		CDD18.3	552	0	0	0	3	38	112	178	169	46	4	0	0	(11)
(12)		CDH23.3	4950	0	0	1	32	321	961	1674	1597	340	24	0	0	(12)
(13)		CDH26.7	1863	0	0	0	6	71	331	687	676	91	2	0	0	(13)
(14)	Wind	WSAvg	2.8	3.1	3.2	3.4	3.1	2.8	2.7	2.5	2.4	2.5	2.6	2.7	3.0	(14)
(15)	Precipitation	PrecAvg	485	29	29	27	39	48	67	48	44	45	33	36	34	(15)
(16)		PrecMax	833	145	95	87	84	134	156	208	145	138	136	111	136	(16)
(17)		PrecMin	363	1	4	1	1	6	6	0	5	1	2	4	0	(17)
(18)		PrecStd	105	26	21	21	19	28	34	42	35	39	33	29	27	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	16.5	21.3	27.2	30.9	34.1	36.4	36.4	32.1	26.2	20.7	13.6	(19)
(20)			MCWB	7.9	10.7	12.0	16.6	18.5	21.7	20.9	21.9	19.2	17.8	15.0	10.5	(20)
(21)		2%	DB	8.6	12.9	18.5	23.4	28.6	31.9	34.0	34.1	29.2	23.3	17.3	11.1	(21)
(22)			MCWB	6.2	8.5	11.0	14.6	18.2	20.8	21.4	21.6	18.4	16.4	13.1	9.0	(22)
(23)		5%	DB	6.6	10.4	15.9	21.0	26.8	30.2	32.2	32.4	27.1	20.9	14.9	8.8	(23)
(24)			MCWB	5.0	7.3	9.7	13.3	17.4	20.2	21.2	20.8	17.9	15.3	12.1	7.0	(24)
(25)		10%	DB	4.9	7.9	13.3	18.9	24.9	28.5	30.4	30.5	25.1	18.6	13.0	6.7	(25)
(26)			MCWB	3.8	5.7	8.2	12.4	16.7	19.5	20.6	20.2	17.2	14.1	11.1	5.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	11.2	13.0	17.2	20.7	23.7	24.5	25.6	21.5	19.4	15.5	11.1	(27)
(28)			MCDWB	10.6	15.1	19.9	25.1	27.2	30.2	31.4	32.4	26.8	24.2	19.3	12.9	(28)
(29)		2%	WB	6.5	9.0	11.4	15.5	19.5	22.3	23.3	23.5	20.2	17.4	13.8	9.2	(29)
(30)			MCDWB	8.3	12.4	17.4	22.1	26.1	28.8	29.8	30.0	24.9	21.2	16.7	11.2	(30)
(31)		5%	WB	5.2	7.5	10.1	14.2	18.4	21.4	22.5	22.2	19.2	16.0	12.5	7.2	(31)
(32)			MCDWB	6.4	9.9	15.1	19.4	24.7	28.0	29.0	28.7	24.2	19.8	14.5	8.6	(32)
(33)		10%	WB	3.8	6.0	8.8	13.0	17.4	20.5	21.7	21.2	18.2	14.7	11.2	5.5	(33)
(34)			MCDWB	4.9	7.9	12.5	17.6	23.1	26.7	28.1	27.7	23.3	18.0	13.0	6.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.4	6.7	8.7	9.8	10.6	11.0	11.1	10.2	8.5	6.4	5.4	(35)	
(36)			MCDBR	7.2	10.6	13.0	13.0	13.1	12.7	13.1	13.5	12.9	11.5	9.2	7.5	(36)
(37)			MCWBR	5.3	6.7	6.8	6.1	5.5	4.9	4.5	4.7	4.8	5.7	5.4	5.5	(37)
(38)		5% WB	MCDWB	6.7	9.6	12.0	11.7	12.0	11.6	11.5	11.8	11.2	10.5	8.0	7.2	(38)
(39)			MCWBR	5.2	6.6	6.8	5.8	5.5	5.1	4.7	4.9	5.0	5.7	5.0	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.339	0.370	0.404	0.456	0.429	0.428	0.446	0.460	0.420	0.387	0.363	0.339	(40)	
(41)		taud	2.289	2.224	2.183	2.056	2.219	2.256	2.213	2.164	2.237	2.311	2.324	2.309	(41)	
(42)		Ebn at Noon	744	790	816	803	842	843	821	792	791	763	711	700	(42)	
(43)		Edh at Noon	82	106	127	157	138	133	137	139	119	96	80	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.25	2.06	3.27	4.63	5.84	6.19	6.24	5.62	4.08	2.60	1.39	1.05	(44)	
(45)		RadStd	0.18	0.27	0.32	0.44	0.42	0.41	0.40	0.34	0.35	0.26	0.17	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.65	N/A	N/A	+0.70	N/A	N/A	-123	-176	+131	+90
(47) Regional (47 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	N/A	-129	+92	+58

Nomenclature: See separate page